

Module 02: Relational Calculus (TRC and DRC)

Relational Calculus: TRC vs DRC Foundations

Declarative Non-Procedural Query Languages, Quantifiers & Codd's Equivalence

Tuple Relational Calculus (TRC)

Filters entire tuples: { t | P(t) }

$$\{ t \mid t \in \text{EMPLOYEE} \wedge t.\text{salary} > 50000 \}$$

"Find all tuples t in EMPLOYEE having salary > 50k"

First-Order Logic Quantifiers:

- Existential Quantifier ($\exists$): $\exists s \in R (P(s))$ - True if at least one exists
- Universal Quantifier ($\forall$): $\forall s \in R (P(s))$ - True if all satisfy P

Equivalence: $\forall s (P(s)) \equiv \neg(\exists s (\neg P(s)))$

Safe TRC Expressions

Query MUST NOT produce infinite tuples!
Unsafe: { t | $\neg(t \in \text{EMPLOYEE})$ } → Returns infinite universe!

Domain Relational Calculus (DRC)

Filters individual attribute domain variables: { (x1..xn) | P }

$$\{ (e, n) \mid \exists d, s ((e, n, d, s) \in \text{EMP} \wedge d = \text{'Sales'}) \}$$

"Get EmpID (e) and Name (n) of employees in Sales dept"

Domain Variables vs Tuples:

- TRC variable represents an entire row/tuple (tname, t.age).
- DRC variables represent specific column values ((id, name, sal)).

Foundation of QBE (Query-By-Example) visual query builders!

Safe DRC Expressions

All values in result must come from the domain of the database.
Guarantees safety across finite relation states.

Codd's Theorem (Relational Completeness Equivalence)

$\text{Relational Algebra} \equiv \text{Safe TRC} \equiv \text{Safe DRC}$
Every query expressible in Relational Algebra can be written in Safe TRC and Safe DRC, forming the bedrock of modern SQL.

Non-Procedural Power: Relational Calculus specifies WHAT data is needed using mathematical First-Order Logic, without specifying the procedural access path.

1. Tuple Relational Calculus (TRC)

Format: { t | P(t) }. Tuple variable t represents an entire row. Quantifiers $\exists$ (There exists) and $\forall$ (For all) allow joins and universal subset queries.

```
-- Find names of employees in 'Research' earning > 60000:
{ t.name | t ∈ EMPLOYEE ∧ t.dept = 'Research' ∧ t.salary > 60000 }
```

2. Domain Relational Calculus (DRC)

Format: { (x1, x2, ...) | P(x1, x2, ...) }. Variables represent individual column domains.

```
-- Find emp_id and name of employees in 'Sales':
{ (e, n) | ∃ d, s ((e, n, d, s) ∈ EMPLOYEE ∧ d = 'Sales') }
```

Safe Queries: Unsafe queries like { t | $\neg(t \in R)$ } return infinite universes. A calculus expression is Safe if all resulting values are bounded within the active domain of the database.

3. Codd's Theorem

Equivalence: Relational Algebra $\equiv$ Safe TRC $\equiv$ Safe DRC. Any database query language matching this power is termed *Relationally Complete*.